

SURGUT, Russian Federation

WMO: 238490

Lat: 61.3424N

Lon: 73.4220E

Elev: 56

StdP: 100.65

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.2	(c) -37.3	(d) -43.8	(e) 0.1	(f) -40.0	(g) -40.9	(h) 0.1	(i) -37.2	(j) 9.8	(k) -15.5	(l) 8.9	(m) -16.8	(n) 2.1	(o) 70	(p) 0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.1	(c) 28.8	(d) 18.6	(e) 26.9	(f) 17.9	(g) 24.9	(h) 17.4	(i) 20.2	(j) 25.4	(k) 19.3	(l) 24.3	(m) 18.3	(n) 23.3	(o) 4.5	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.5	22.1	17.2	12.4	21.5	16.1	11.5	20.8	58.5	25.2	55.1	24.0	51.9	23.3	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.7	(c) 7.7	(e) -42.4	(f) 31.3	(g) 4.4	(h) 1.4	(i) -45.5	(j) 32.3	(k) -48.1	(l) 33.2	(m) -50.6	(n) 33.9	(o) -53.8	(p) 35.0		
(a) 9.9	(b) 8.7	(c) 7.7	(e) -42.0	(f) 21.7	(g) 3.7	(h) 1.2	(i) -44.6	(j) 22.5	(k) -46.8	(l) 23.2	(m) -48.8	(n) 23.8	(o) -51.5	(p) 24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-20.8	-17.9	-9.0	-1.9	6.2	14.7	17.7	14.3	7.8	0.0	-12.2	-17.7	
	DBStd	14.97	9.45	8.67	7.13	6.92	6.22	5.49	4.15	4.00	4.45	5.64	8.89	9.39	
	HDD10.0	4796	956	782	590	357	154	19	1	7	95	311	666	860	
	HDD18.3	7314	1214	1016	848	606	379	134	64	137	316	568	916	1118	
	CDD10.0	609	0	0	0	1	36	161	240	139	30	2	0	0	
	CDD18.3	86	0	0	0	0	3	26	44	12	1	0	0	0	
	CDH23.3	736	0	0	0	0	35	243	376	80	2	0	0	0	
	CDH26.7	168	0	0	0	0	6	59	91	11	0	0	0	0	
Wind	WSAvg	3.9	3.5	3.5	4.2	4.4	4.4	4.0	3.6	3.6	3.8	4.1	3.9	3.6	
Precipitation	PrecAvg	824	53	43	59	54	64	78	71	89	74	86	85	67	
	PrecMax	1031	102	83	151	85	150	137	170	151	146	150	138	135	
	PrecMin	652	15	11	3	2	33	35	30	34	29	37	31	27	
	PrecStd	97	22	17	28	20	27	23	29	28	27	29	27	23	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.1	5.6	16.6	27.1	30.8	31.1	28.1	22.8	14.8	4.2	0.2	
		MCWB	-0.1	0.4	1.6	9.7	15.2	17.9	19.8	20.3	15.8	10.5	3.6	-0.2	
	2%	DB	-3.1	-1.8	3.0	12.2	23.2	28.2	29.1	25.8	19.6	11.2	2.1	-1.2	
		MCWB	-3.4	-2.3	0.3	7.0	13.5	17.6	19.1	18.9	14.3	8.5	1.6	-1.5	
	5%	DB	-5.8	-4.1	1.2	9.2	19.9	26.2	27.2	23.2	17.0	9.2	0.9	-3.0	
		MCWB	-6.3	-4.6	-0.4	5.1	12.5	17.2	18.4	17.3	12.8	7.3	0.5	-3.3	
	10%	DB	-8.0	-6.2	0.0	7.1	16.1	23.9	25.2	21.1	14.8	7.4	-0.3	-5.2	
		MCWB	-8.4	-6.8	-1.3	3.9	10.4	16.4	17.8	16.4	11.9	5.8	-0.7	-5.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.6	2.0	10.2	16.2	20.6	21.6	21.3	16.7	11.0	3.6	0.0
			MCDB	0.1	1.2	4.1	15.7	24.1	25.8	27.0	26.1	21.1	13.4	4.0	0.2
2%		WB	-3.5	-2.3	0.8	7.7	14.3	19.2	20.7	19.6	14.9	9.3	1.5	-1.7	
		MCDB	-3.3	-1.9	2.2	11.5	21.3	25.0	26.1	23.8	18.5	11.1	1.8	-1.5	
5%		WB	-6.3	-4.7	-0.1	5.6	12.7	18.3	19.7	18.4	13.3	7.4	0.3	-3.4	
		MCDB	-5.9	-4.3	1.1	8.6	18.8	24.1	24.8	22.1	16.1	9.0	0.6	-3.1	
10%		WB	-8.5	-6.9	-1.4	4.0	10.9	17.1	18.7	17.1	11.9	5.8	-1.0	-5.6	
		MCDB	-8.2	-6.3	-0.1	6.8	16.2	23.0	23.9	20.3	14.5	7.2	-0.5	-5.3	
Mean Daily Temperature Range		5% DB	MDBR	7.4	8.1	9.5	9.2	9.5	9.4	9.1	8.0	6.9	5.4	6.6	7.3
			MCDBR	8.0	7.2	8.3	10.8	13.9	12.7	11.8	11.2	10.0	7.2	4.8	7.7
	MCWBR		8.0	7.0	6.6	6.8	6.7	5.3	4.7	5.6	5.6	4.9	4.7	7.6	
	5% WB	MCDBR	8.1	7.3	7.9	9.9	13.1	11.2	10.4	10.1	9.2	6.8	5.2	7.7	
		MCWBR	8.1	7.1	6.7	6.9	7.2	5.5	4.9	5.8	5.8	5.0	5.2	7.6	
Clear-Sky Solar Irradiance	taub	0.254	0.298	0.351	0.373	0.386	0.400	0.421	0.414	0.357	0.327	0.264	0.237		
	taud	1.960	2.011	1.958	2.072	2.291	2.345	2.277	2.264	2.387	2.238	2.020	1.899		
	Ebn at Noon	420	639	729	805	833	827	794	764	750	616	402	257		
	Edh at Noon	43	80	121	132	115	112	117	109	81	65	40	27		
All-Sky Solar Radiation	RadAvg	0.36	1.23	2.88	4.54	5.05	5.38	5.12	3.60	2.18	0.94	0.36	0.17		
	RadStd	0.03	0.12	0.16	0.46	0.32	0.43	0.53	0.38	0.23	0.10	0.08	0.01		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-368	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-368	N/A	N/A

Nomenclature: See separate page